

# Microwave Control-Chain Spoofing of NV Magnetometers: Detection Limits and Reference-Sensor Design Rules

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**Abstract**—Diamond magnetometers based on nitrogen-vacancy centers are moving from the laboratory to field applications such as magnetic anomaly navigation, where a wrong field estimate becomes a wrong position. The quantum sensor is hard to reach remotely, but its classical microwave control chain is not. We ask whether the sensor can detect manipulation of its own drive, whether an auxiliary reference sensor can restore a correct estimate, and which hardware choices decide the outcome. Offsetting the microwave drive frequency by less than one part per million reproduces the photon statistics of a genuine field change for any Ramsey interrogation schedule, so the sensor cannot verify its own output; secret interrogation timing stops only precomputed phase attacks, and driving both spin transitions analytically reduces a clock error’s bias tenfold. Falling back to a reference sensor on alarm cannot push the rate of harmful outputs below the reference’s own error rate, and a closed-form design equation predicts a blind window of small spoofs that pass unnoticed, which fixes how precise the reference must be. In a simulation study with 30 independently seeded pipelines and a protocol fixed before the run, the equation matches all eight tested reference conditions within one percentage point, and a reference that shares the primary sensor’s clock releases a harmful estimate on almost every attack.

**Index Terms**—NV magnetometry, quantum sensor security, sensor spoofing, microwave control, reference sensor fusion, Ramsey magnetometry

## I. INTRODUCTION

Nitrogen-vacancy (NV) magnetometers combine high sensitivity with room-temperature operation, which makes them candidates for field deployment [1]. One such application is magnetic anomaly navigation, where the measured field is matched against a map to obtain position [2]. In this setting a field error is a navigation error, so the integrity of the released estimate matters as much as its precision. The question is where an attacker could corrupt it.

The weak point is not the quantum spin but the electronics that control it. An NV Ramsey magnetometer reads the field as a phase that accumulates between two microwave pulses. That phase is measured against a reference set by classical hardware: a reference clock, a microwave synthesizer, an arbitrary waveform generator (AWG), and the cables between them. Signal injection into the analog front ends of sensors is well documented [3], [4]. The manipulation an NV sensor needs is also small. A 2.8 kHz drive offset, 0.89 ppm of the 3.15 GHz transition, imitates a 100 nT field change. Every tone derived from a shifted clock inherits the clock’s fractional error, so pulling the usual 10 MHz reference by about 10 Hz, or rewriting the synthesizer’s frequency setting, is enough. This raises a first question: can the sensor notice?

Existing defenses cannot answer this, because none was designed for an attacker on the drive. NV sensing research treats control errors as benign imperfections to be averaged out by robust or double-quantum sequences [5]–[7]. Spoofing defenses for classical active sensors hide probing times from the attacker [8], a strategy that fast attackers can bypass [9]. A common engineering answer is to add a second, independent sensor and fall back to it when the two disagree. Such detectors are usually judged by how often they raise an alarm,

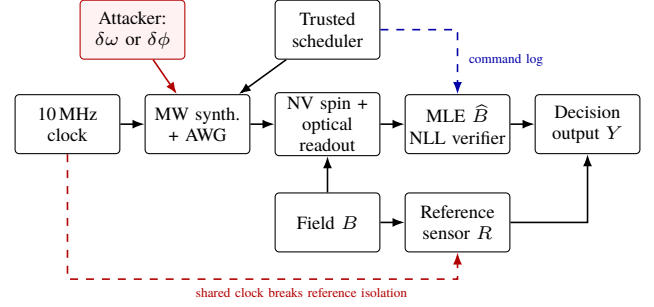


Fig. 1. Sensor chain and trust boundaries. The attacker offsets the drive frequency ( $\delta\omega$ ) or analysis phase ( $\delta\phi$ ) after the trusted scheduler; the command log stays protected. The reference sensor is secure only if it shares no clock, synthesizer or data path with the primary chain (red dashed path).

yet for a field estimate what matters is whether the value released after the alarm is correct. We therefore ask: *when an attacker manipulates the microwave drive, what can the defender detect, how much can a reference sensor recover, and which hardware choices decide the outcome?* We make three contributions:

- **Control-chain attack surface** (Section III). A drive-frequency offset is invisible in the photon statistics; secret timing stops only precomputed phase attacks; dual-transition driving cancels common clock errors.
- **Reference architecture and design equation** (Section IV). Fallback cannot beat the reference’s own error rate under attack, and a closed-form equation predicts residual harm and sizes the reference.
- **Engineering evaluation and design rules** (Sections V–VI). A 30-pipeline simulation confirms both results, prices a shared clock, and maps each countermeasure to the hardware it requires.

## II. SENSOR CHAIN AND THREAT MODEL

The attacker’s lever is the part of the chain that turns commands into microwave pulses (Fig. 1). A trusted scheduler chooses the interrogation time  $t_k$  and analysis phase  $\phi_k$  of each measurement setting  $k$  and records them in a protected command log. The synthesizer and AWG turn these commands into pulses on one NV transition, the optical readout returns photon counts, and the estimator computes a field estimate  $\hat{B}$  from the counts and the logged commands. A verifier scores settings held out from the fit, and an optional reference sensor supplies an independent reading  $R$  of the same field. The decision stage combines them into the released output  $Y$ .

The attacker acts between the scheduler and the spin. It can add a constant offset to the drive frequency, or change the analysis phase of each pulse, either from a table prepared in advance or with real-time knowledge of the schedule. It cannot observe the true field or the

reference, and it cannot alter counts, logs, software or the reference path. An attacker with that access would control the output directly, a threat that calls for data-path protection rather than the measurement-level checks studied here.

We judge a defense by what it releases, not by how often it alarms. A release  $Y$  is *harmful* if it misses the true field by more than  $\epsilon_H = 25$  nT, and the attacker aims to shift the estimate by 100 nT without an alarm. Both values are evaluation parameters, not operational standards. Because a defense can avoid harm trivially by releasing nothing, we also report *coverage*, the fraction of attempts that release an output, and the root-mean-square error (RMSE) of released outputs. With these measures, we can ask what the primary sensor can detect on its own.

### III. CAN THE SENSOR DETECT THE ATTACK ON ITS OWN?

A Ramsey sequence turns the field into a phase and the phase into a spin population. The first microwave pulse creates a superposition, and the spin then precesses for a time  $t$  at a rate set by its detuning from the drive. The second pulse, with analysis phase  $\phi$ , converts the accumulated phase into a population that the optical readout measures. The probability of reading the  $m_s = 0$  state is [10], [11]

$$p = \frac{1}{2} [1 + (1 - 2\eta) c e^{-t/T_2^*} \cos \Theta], \quad \Theta = (\gamma B + \delta\omega)t + \beta - \phi, \quad (1)$$

where  $\gamma$  is the gyromagnetic ratio,  $B$  the field and  $\delta\omega$  any detuning of the drive. The preparation phase is  $\beta$ , the contrast  $c$ , the dephasing time  $T_2^*$ , and  $\eta$  a symmetric readout error. Repeating each setting  $N$  times gives binomially distributed photon counts.

**Frequency offsets are undetectable.** Equation (1) depends on the field and the drive only through their sum  $\gamma B + \delta\omega$ . An attacker who lowers the drive frequency by  $\gamma\Delta B$  therefore produces exactly the count distribution of a field  $B + \Delta B$ . This holds for any pulse shape and any schedule, including secret ones, as long as the dynamics of the addressed transition depend on the field only through this detuning. Because the clean and attacked data have the same distribution, every test computed from the primary sensor's data raises an alarm with the same probability in both cases. In the terms of cyber-physical security, the attack is *undetectable*: its measurements are exactly those of a legitimate state [12].

The equivalence is exact only in the two-level model, so we checked the neglected physics. Equation (1) ignores the second, unaddressed NV transition, the counter-rotating part of the drive, and the  $^{14}\text{N}$  nuclear spin. We simulated the full coupled electron-nuclear system in the laboratory frame, with finite 25 ns pulses and no rotating-wave approximation, at eight random settings per bias field. The largest difference between the populations under a real field change and under the equivalent drive offset was  $1.1 \times 10^{-4}$  at a 10 mT bias field and  $4.8 \times 10^{-4}$  at 5 mT. For comparison, the spoof itself changes the population by a median of 0.034, and shot noise at 1000 shots is 0.016 per setting. If the residual stays at this level, even an ideal detector that knew it exactly would need at least 5,493 sequences at 10 mT (293 at 5 mT) to detect it with 50% probability at a 1% false-alarm rate. The attack is therefore undetectable within any single sequence. Can the defender do better by hiding the schedule?

**Secret timing stops only precomputed phase attacks.** A phase actuator imitates a field change only if it applies  $\delta\phi_k = -\gamma\Delta B t_k$  with the true interrogation time. If the defender draws each  $t_k$  secretly, an attacker that committed to a table built for public nominal times  $t_{\text{nom}}$  leaves a residual phase  $\gamma\Delta B (t_{\text{nom}} - t_k)$  that varies from setting to setting. No single field value then explains all settings, and a likelihood test on held-out settings detects the misfit. A frequency

offset needs no knowledge of  $t_k$ , because the extra phase grows with time automatically. Secret timing therefore does not help against it.

**Dual-transition driving cancels common clock errors.** The NV ground state offers a hardware defense. It has two microwave transitions,  $m_s = 0 \rightarrow +1$  and  $m_s = 0 \rightarrow -1$ , whose frequencies move in opposite directions when the field changes:  $\omega_{\pm} = D \pm \gamma(B_0 + B)$ , where  $D \approx 2.87$  GHz is the zero-field splitting and  $B_0$  the bias field. A clock error, by contrast, shifts both drive tones in the same direction. Differencing the two measured detunings keeps  $2\gamma(B_0 + B)$  and cancels offsets common to both tones, including drifts of  $D$  [6], [7], [13]. A fractional clock error  $\epsilon$  scales both tones. At  $\epsilon = 1$  ppm and  $B_0 = 10$  mT it biases a single-transition estimate by about 112 nT but a dual-transition estimate by only 10 nT. To spoof the dual-transition sensor, the attacker must inject a precise differential offset on two tones. This is an algebraic capability bound, not a demonstrated defense. A single-transition sensor therefore needs help from outside the chain; how much can an external reference provide?

### IV. CAN A REFERENCE SENSOR RESTORE THE ESTIMATE?

**Two checks raise the alarm.** Each sequence splits its settings into an estimation set, on which the maximum-likelihood estimate  $\hat{B}$  is fitted, and a verification set. The first check,  $S_N$ , is the negative log-likelihood of the verification counts under  $\hat{B}$ ; it grows when the data do not fit a single field. The second check,  $S_R = |\hat{B} - R|$ , compares the primary estimate with the reference reading. The defender first fixes a false-alarm budget  $\alpha$ , the fraction of clean sequences allowed to raise an alarm; we use  $\alpha = 1\%$ . Each check receives half of the budget: its threshold is set so that only a fraction  $\alpha/2$  of its scores on clean calibration sequences lie above it [14]. The alarm fires when either check exceeds its threshold, so on clean data it fires on at most  $\alpha$  of sequences. We call the discrepancy threshold  $\kappa$ .

**The reference is only useful if it is isolated.** Ideally, the reference measures the same field through its own electronics, so the attack cannot reach it. In practice, it may share a clock or synthesizer with the primary sensor, or carry an uncorrected calibration offset. This matters because whatever the reference shares with the primary chain, the attacker can shift too. We model the reading as

$$R = B + b_R + \lambda\Delta B + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_R^2), \quad (2)$$

where  $\sigma_R$  is the reference noise,  $b_R$  a calibration offset, and  $\lambda$  the fraction of the spoof  $\Delta B$  that also reaches the reference. An independent reference has  $b_R = \lambda = 0$ . A second NV sensor driven from the same clock has  $\lambda = 1$ .

**Output policies.** Once the alarm is decided, the decision stage chooses what to release. *Fallback* releases  $R$  on alarm and  $\hat{B}$  otherwise. *Abstention* releases  $\hat{B}$  only without an alarm and releases nothing otherwise. *Always-reference* ignores the primary sensor. *Fusion* rules blend the two sensors instead of switching. Inverse-variance fusion releases  $R + w(\hat{B} - R)$  with  $w = \sigma_R^2 / (\sigma_R^2 + \sigma_P^2)$ , weighting each sensor by its precision, where  $\sigma_P$  is the primary sensor's clean error. Gated fusion uses fallback on alarm and inverse-variance fusion otherwise. Clipped fusion releases  $R + \text{clip}(\hat{B} - R, \pm\kappa)$ , so it follows the primary sensor only within the alarm threshold. Which of these policies limits harm under attack?

**The reference sets a floor.** A fallback release is harmless only if the released sensor is accurate. When the spoof makes  $\hat{B}$  harmful, only a released, accurate  $R$  can avoid harm, so fallback harm is at least the reference's own harm rate, whatever the alarm rule. For an independent reference with Gaussian noise, that rate is  $2\Phi(-\epsilon_H/\sigma_R)$ , where  $\Phi$  is the standard normal cumulative distribution function. This is the probability that the reference alone misses the true field by

more than  $\epsilon_H$ . The same floor limits every fusion rule of the form  $Y = R + g(\hat{B} - R)$ , which includes all rules above. If the attacker may choose the spoof size freely,  $\hat{B}$  carries no usable information about  $B$ , and worst-case harm cannot fall below the floor. A floor is only a lower bound, though. How close does fallback get to it?

**Design equation.** Assume the spoof shifts the primary estimate by exactly  $d > h$ , with  $h = \epsilon_H$ , and neglect the primary sensor's own noise, which is small next to the reference's. The reference then reads  $R = B + s + \varepsilon$  with shift  $s = b_R + \lambda d$ . Harm arises in two disjoint ways, which give the two terms of

$$H_F(d) = P(|s + \varepsilon| > h) + P(|s + \varepsilon| \leq h, |d - s - \varepsilon| \leq \kappa). \quad (3)$$

The first term covers an inaccurate reference: the released value is harmful whether or not the alarm fires, because both sensors are wrong. The second term covers an accurate reference that still leaves the discrepancy  $|\hat{B} - R|$  below  $\kappa$ . No alarm fires, so the spoofed  $\hat{B}$  is released.

For an independent reference ( $s = 0$ ), the discrepancy is close to  $d$ , and the second term separates three regimes of spoof size:

- *Blind window*,  $h < d \leq \kappa - h$ . The spoof is large enough to be harmful but too small to push an accurate reference past the threshold. Every accurate reference is silent, so harm is 1.
- *Transition region*,  $\kappa - h < d < \kappa + h$ . The alarm fires for some reference noise values and not others, so harm falls as  $d$  grows. This region matters because moderate spoofs land here and are only partly caught.
- *Floor*,  $d \geq \kappa + h$ . Every accurate reference raises the alarm, so only the first term remains.

The threshold follows from the reference and primary noise. Because the discrepancy can deviate in either direction, its check is two-sided: the budget  $\alpha/2$  is split equally between large positive and large negative discrepancies. This gives

$$\kappa \approx z_{1-\alpha/4} \sqrt{\sigma_R^2 + \sigma_P^2}, \quad (4)$$

where  $z_{1-\alpha/4} = 2.807$  is the standard normal quantile. The blind window exists whenever  $\kappa > 2h$ . With the simulated primary noise  $\sigma_P \approx 4.28$  nT, this happens once  $\sigma_R > 17.3$  nT. At  $\sigma_R = 20$  nT, spoofs from 25 to 32.6 nT pass unnoticed. Equations (3) and (4) thus become a sizing rule for the reference. The next section tests them.

## V. EVALUATION

**Data sources.** Every result comes from a single simulation run whose cases, policies, endpoints and random seeds were fixed and time-stamped before it ran, which we call the frozen run. Two analyses were added afterwards and are marked post hoc. One is the comparison of Eq. (3) with the observations in Fig. 2. The other is the fusion replay in Table III, which evaluates additional output rules on the stored per-attempt records of the frozen run without new simulation. The two-level equivalence check, the spin-1 simulation and the clock arithmetic of Sections I–III are separate calculations.

**Setup.** The run simulates the model of Eq. (1) with  $T_2^* = 100 \mu s$ ,  $\eta = 0.1$ , two analysis phases, and 1000 shots per setting, for 20 settings (20,000 shots) per sequence. Fields are drawn uniformly from  $[0, 1.9] \mu T$ . The defender uses one of three schedules, each evaluated as a separate case so that the benefit of secrecy can be isolated. The first is a fixed public schedule of five nominal times from 5 to 60  $\mu s$ . The second adds random analysis phases. The third draws secret times uniformly from 5 to 60  $\mu s$  at every setting. The run contains 20 cases. Attack cases vary the actuator (phase or frequency), the schedule, and the attacker's knowledge of the actual times. Reference-stress cases

TABLE I  
ATTACK CAPABILITY AGAINST THE LIKELIHOOD CHECK ALONE (FROZEN RUN, 15,360 ATTACKED ATTEMPTS PER ROW). TARGET: ESTIMATE SHIFTED BY 100 nT. UNDETECTED: TARGET REACHED WITHOUT ALARM. DETUNING: DRIVE-FREQUENCY OFFSET.

| Attack / defender schedule         | Target | Undetected |
|------------------------------------|--------|------------|
| Nominal phase / fixed              | 14,879 | 14,758     |
| Nominal phase / hidden phase       | 14,669 | 14,593     |
| Nominal phase / hidden time        | 2,569  | 1          |
| Late disclosure / hidden time      | 2,569  | 1          |
| Timely disclosure / hidden time    | 15,004 | 14,892     |
| Full knowledge / hidden time       | 15,004 | 14,892     |
| Detuning / hidden time             | 15,004 | 14,892     |
| Selected phase table / hidden time | 6,081  | 17         |
| Mean-time phase / hidden time      | 5,900  | 14         |
| Zero offset / hidden time          | 0      | 0          |

vary  $\sigma_R \in \{5, 15, 50\}$  nT, add a 10 nT offset or a shared fraction  $\lambda \in \{0.5, 1\}$ , or shrink the spoof to 50 nT. Control cases include no attack, a genuine 100 nT field change, and uncalibrated readout drift. Thirty independently seeded pipelines each calibrate their own thresholds and attacker tables and simulate 512 clean/attacked pairs per case, giving 15,360 attempts per case and condition. Intervals are 95% bootstrap intervals over the 30 pipelines. Unless stated otherwise, results refer to the *nominal configuration*: a 100 nT frequency spoof,  $\sigma_R = 20$  nT, and thresholds calibrated at  $\alpha = 1\%$ .

### Secret timing stops phase tables but not frequency offsets.

Table I reports how often each attack reaches its 100 nT target without an alarm, using only the likelihood check. Each row pairs an attack with the defender's schedule. In the disclosure rows, the attacker learns each actual time either before the  $2 \mu s$  its actuator needs to apply the phase (timely) or after it (late); full knowledge gives the time immediately. Mean-time and zero-offset tables are simple baselines for the optimized table. A nominal-time phase attack succeeds on 14,758 of 15,360 attempts against the fixed schedule but on only 1 against secret times. Randomizing the analysis phase alone does not help, because the attacker simply adds its offset to whatever phase is used. The attacker's strongest option against secret times is a precommitted phase table: one phase offset per public nominal-time slot, chosen by numerical optimization (differential evolution) on simulated data to shift the estimate while still fitting the verifier. It shifts the estimate in 6,081 attempts but passes the verifier in only 17; since the search is finite, this is a lower bound on what a phase attacker can do. A drive-frequency offset needs no schedule knowledge and succeeds on 14,892 of 15,360 attempts even against secret times, with simulated count distributions matching the field-shifted ones to within  $5.55 \times 10^{-16}$ . With the primary sensor blind to this attack, everything rests on the reference.

### Fallback reaches the floor, and the design equation predicts it.

In the nominal configuration, fallback releases a harmful estimate on 20.78% [20.26, 21.30] of attacks, the same rate as always-reference. This is the reference's own error rate at  $\sigma_R = 20$  nT, so no alarm rule can remove it. The design equation reproduces this and every other tested condition without fitting to attack data. Evaluated with the calibrated thresholds, it agrees with the observations within 0.60 percentage points in all 8 reference conditions of the setup (Fig. 2). The equation also explains the failures. A 50 nT spoof falls in the transition region and raises harm to 74.93%. A noisier reference with  $\sigma_R = 50$  nT widens  $\kappa$  until even the 100 nT spoof sits in the blind window, and fallback harm rises to 99.76%, worse than the reference alone (62.25%). A precise reference with  $\sigma_R = 5$  nT removes harm

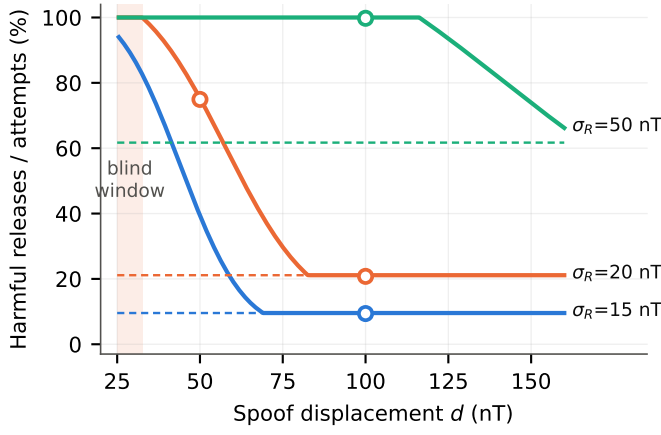


Fig. 2. Fallback harm versus spoof size predicted by Eq. (3) (post hoc) for three reference noise levels (lines), their reference floors (dashed), and frozen-run observations with 95% intervals (circles). Shading: blind window at  $\sigma_R = 20$  nT, spoofs from 25 to 32.6 nT that are harmful yet too small to trigger the discrepancy alarm.

TABLE II  
REFERENCE DEPENDENCY AUDIT (FROZEN RUN): HARMFUL RELEASES OF FALLBACK UNDER A 100 nT SPOOF,  $\sigma_R = 20$  nT.

| Shared dependency           | Model effect                       | Harm   |
|-----------------------------|------------------------------------|--------|
| None (independent)          | $b_R = \lambda = 0$                | 20.78% |
| Calibration or offset error | $b_R = 10$ nT                      | 26.94% |
| Partially coupled paths     | $\lambda = 0.5$                    | 89.67% |
| Shared clock or synthesizer | $\lambda = 1$                      | 99.98% |
| Shared data path/software   | attacker rewrites $R$ out of scope |        |

entirely. Reference precision is one hardware requirement; isolation is the other.

**A shared clock defeats the reference.** Each departure from isolation in Eq. (2) raises harm, as Table II shows. A 10 nT reference offset raises it to 26.94%, and partial coupling ( $\lambda = 0.5$ ) to 89.67%. A reference that inherits the full spoof, as a second NV sensor on the same clock would, releases harmful estimates on 99.98% of attacks. Its alarm fires on only 0.94% of them, because both sensors move together and never disagree.

**What fallback, abstention and fusion buy.** If fallback cannot beat the reference under attack, why use the primary sensor at all? The answer is clean precision. Without attack, fallback releases the primary estimate most of the time, and its RMSE is 6.14 nT, against 19.98 nT for always-reference. The paired difference is -13.84 nT, with an interval of [-14.27, -13.44] adjusted for multiple comparisons. This precision costs 20,000 primary shots per estimate. The fusion replay in Table III shows the price of pushing precision further. Inverse-variance fusion is the most precise rule on clean data, but it trusts the precise, compromised primary sensor and releases a harmful estimate on every attack. Gated and clipped fusion fall in between, and every rule except always-reference has a worst case far above the reference floor, which is only a lower bound.

Abstention trades availability for harm. It lowers harm by 18.85 percentage points relative to fallback, but only by releasing almost nothing: coverage under attack drops by 98.07 points. All 296 outputs it still releases are harmful, and each release costs 1.04 million primary shots. Without attack, the alarm fires on 1.07% [0.91, 1.24] of sequences, close to the 1% budget. A genuine 100 nT field change raises only 154 alarms in 15,360 attempts, because both sensors see

TABLE III  
OUTPUT RULES REPLAYED POST HOC ON THE FROZEN PER-ATTEMPT RECORDS,  $\sigma_R = 20$  nT (RULES DEFINED IN SECTION IV). WORST CASE: MODEL MAXIMUM OVER SPOOFS OF 0–300 nT.

| Rule                    | Clean RMSE (nT) | Attacked harm (%) | Worst case (%) |
|-------------------------|-----------------|-------------------|----------------|
|                         |                 | 100 nT            | 50 nT          |
| Always reference        | 19.98           | 20.78             | 20.78          |
| Combined fallback       | 6.14            | 20.78             | 74.93          |
| Gated fusion            | 6.09            | 20.78             | 74.93          |
| Inverse-variance fusion | 4.29            | 100.00            | 100.00         |
| Clipped fusion          | 4.38            | 94.86             | 94.86          |

TABLE IV  
COUNTERMEASURES, WHAT THEY STOP, AND THEIR HARDWARE COST (FROM SECTIONS III–V).

| Countermeasure                      | Stops                                                                  | Hardware cost                                                                |
|-------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Secret interrogation timing         | Precomputed phase attacks (1 of 15,360 succeed); not frequency offsets | Randomized scheduler; no added hardware                                      |
| Dual-transition drive               | Common clock and synthesizer errors (112 to 10 nT per ppm, analytic)   | Second tone: extra synthesizer channel or IQ sideband                        |
| Independent reference with fallback | Frequency offsets, down to the reference's own error rate              | Second sensor with its own clock; $\sigma_R < 17.3$ nT for a 25 nT tolerance |
| Dependency audit                    | Shared-clock failure (99.98% harm otherwise)                           | Separate clock, synthesizer and data path                                    |

it.

Averaging the reference is a cheaper route to precision if its errors are independent: 11 readings give  $\sigma_R = 6.03$  nT, matching fallback's clean precision with a floor of 0.003%. Fallback pays off only when primary shots are much cheaper than reference readings or drift prevents averaging.

## VI. DESIGN RULES AND LIMITATIONS

The results reduce to four rules for sensor designers, listed with their hardware cost in Table IV. Match the defense to the actuator, since secret timing stops phase tables but not frequency offsets. Size the reference to the application's tolerance rather than to the alarm rate. Audit the reference for shared clocks, synthesizers and data paths. Prefer drive schemes that address both spin transitions.

These rules rest on simulation. The reference is an abstract Gaussian channel without drift, bandwidth or alignment effects. The attacker is modeled as frequency and phase offsets rather than as a circuit-level synthesizer or clock. The attacker's phase-table search is finite. The 30 pipelines replicate sampling and calibration variance within one code base.

## VII. CONCLUSION

A single-transition NV Ramsey magnetometer cannot tell a sub-ppm drive offset from a real field change, so its integrity depends on hardware outside the spin. A reference sensor restores it only down to the reference's own error rate, only if it is precise enough to close the blind window, and only if it shares no clock with the primary sensor. Future work will model the clock and synthesizer at circuit level, simulate two-tone attacks on dual-transition sensors, and test both defenses on NV hardware.

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